

Real Rates, Asset Prices, and Household Entry Barriers

A Public-Data Accounting Note on Payment Affordability

Arne Kotte

AUTHOR

Arne Kotte

DOCUMENT TYPE

Descriptive public-data accounting
note

DATE

May 2026

A Public-Data Accounting Note on Payment Affordability

Author: Arne Kotte **Date:** May 2026 **Document type:** Descriptive public-data accounting note

Abstract

This note asks whether higher real rates restore household entry affordability or shift the barrier from asset prices to financing costs. It is descriptive, not causal. The revised version uses a coherent nominal payment-to-nominal-income affordability screen, states the mortgage-payment identity explicitly, and adds order-invariant accounting, counterfactual rate and price screens, loan-to-value sensitivity, and illustrative cash-to-close and ownership-cost approximations. Public data show two distinct phases. From 2019 to 2021, real rates and mortgage rates fell while home prices rose; the simplified 80 percent loan-to-value payment screen worsened only modestly. From 2021 to 2024, real rates and mortgage rates rose while prices remained elevated; the estimated monthly principal-and-interest payment rose by about \$883 and the payment-to-monthly-income screen rose by 9.28 percentage points. Relative to 2019, the 2024 screen shows a 1.54 percentage-point increase in the 10-year TIPS real yield, a 2.79-point increase in the 30-year mortgage rate, a \$98,725 increase in median sale price, and a roughly \$954 monthly payment increase. The disciplined conclusion is that, in this national public-data screen, observed higher rates did not coincide with enough price adjustment to restore 2019-style entry-payment affordability.

I. Introduction

A common macro story says that low real rates inflate asset prices, while higher real rates discipline valuations and eventually restore affordability. That story can be true in asset-pricing terms and still incomplete for households. A household does not buy a price index. It faces an entry payment, a required down payment, a rent alternative, total cash to close, credit underwriting, insurance and tax costs, and an income constraint.

This note asks a narrow question: **in a transparent public-data screen, did the move to higher real rates coincide with restored household entry affordability, or with a shift from asset-price pressure to financing-cost pressure?** The answer matters because entry barriers shape homeownership access, mobility, family formation, and long-horizon saving capacity.

The paper is a descriptive accounting note. It does not identify the causal effect of real rates on home prices or household formation. It does not estimate a structural housing model. It instead builds a reproducible channel map connecting real rates, mortgage rates, asset prices, nominal income, and representative entry-payment burden. The contribution is measurement discipline and transparent accounting, not a new causal estimate.

II. Conceptual frame

Household entry affordability depends on at least five moving pieces.

First, the **real-rate channel** affects discount rates and asset valuation. Lower real rates can support higher asset prices; higher real rates can pressure valuations. Second, the **mortgage-rate channel** translates financial conditions into monthly payment costs. Third, the **asset-price channel** determines the principal amount that must be financed and the down payment that must be accumulated. Fourth, the **income channel** determines whether the payment is absorbable. Fifth, the **cash-to-close channel** determines whether a household can enter even if the monthly payment appears manageable.

The simplified principal-and-interest burden is:

$$\text{Payment burden} = \frac{[LTV \times \text{Price} \times A(r)]}{[\text{Income} / 12]},$$

where the 30-year monthly annuity factor is:

$$A(r) = \frac{i}{(1 - (1 + i)^{-360})}, \text{ with } i = \text{annual mortgage rate} / 12.$$

This identity is not a causal model, but it is useful measurement discipline. It makes clear that payment burden can worsen because prices rise, mortgage rates rise, income lags, or down-payment constraints force higher loan-to-value borrowing.

III. Data and construction

The note uses public FRED series: the 10-year TIPS real yield (**DFII10**), the 30-year fixed mortgage rate (**MORTGAGE30US**), the Case-Shiller U.S. National Home Price Index (**CSUSHPISA**), median sales price of houses sold (**MSPUS**), nominal median household income (**MEH0INUSA646N**), real median household income (**MEH0INUSA672N**), rent of primary residence CPI (**CUUR0000SEHA**), and real usual weekly earnings (**LES1252881600Q**). Monthly, weekly, and quarterly series are converted to calendar-year annual averages. Annual income data are used as reported.

The main entry-payment screen uses nominal payments divided by nominal income. It assumes an 80 percent loan-to-value mortgage on the median sales price of houses sold. It uses a standard 30-year fixed-payment formula and excludes taxes, insurance, PMI, maintenance, transaction costs, credit constraints, and down-payment accumulation. Later tables add transparent approximations for cash to close and PITI-style monthly costs, but the benchmark measure remains a screen rather than a full affordability model.

The national median home and national median household are not the marginal entrant. Many median households are already owners, while many potential entrants are younger renters with lower liquid wealth and local housing-market exposure. The national screen is therefore a starting point for measurement, not a first-time-buyer estimate. A publishable empirical version should replace this representative screen with renter-specific, starter-home, and local-market measures.

IV. Real rates and mortgage rates moved in different affordability regimes

From 2019 to 2021, the 10-year real yield fell and mortgage rates declined. Asset prices rose quickly, but low financing costs muted the representative payment increase. From 2021 to 2024, the opposite happened: real yields rose, mortgage rates rose, and home prices remained high enough that financing costs dominated the entry-payment screen.

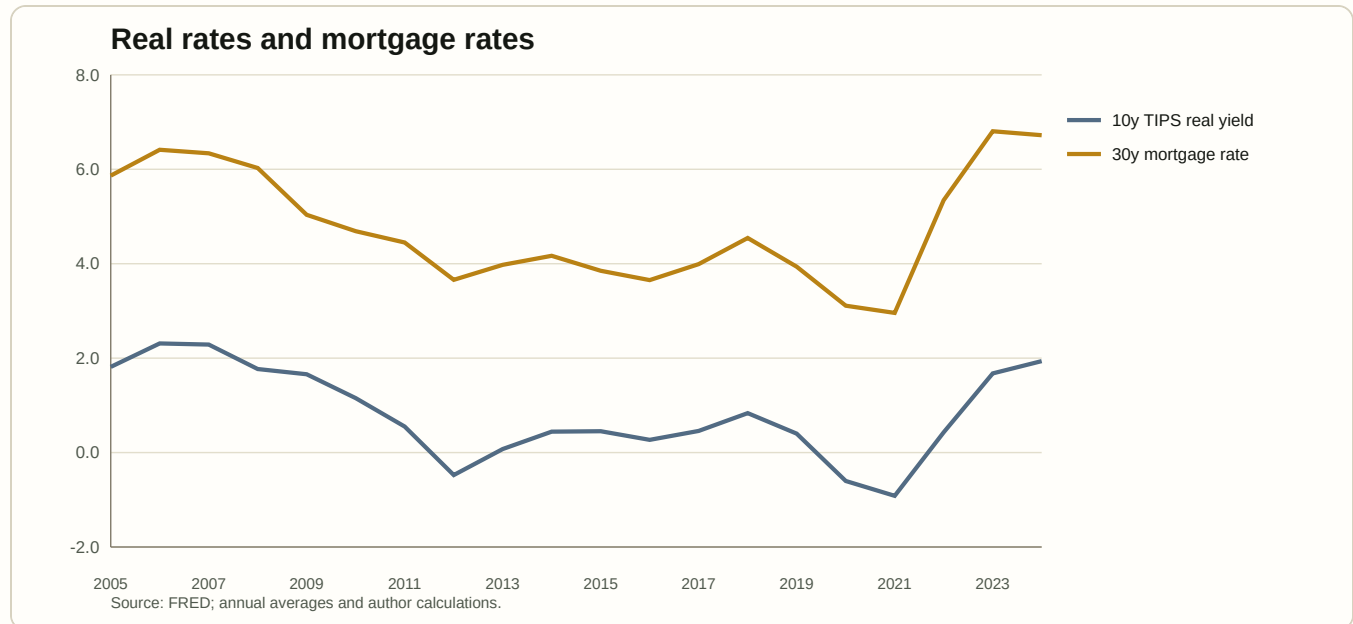


Figure 1. Real rates and mortgage rates.

Table 1. Real-rate, asset-price, and entry-payment indicators. Annual averages. Dollar payment and income entries are nominal. The mortgage-payment screen assumes an 80 percent loan-to-value mortgage on the median sales price of houses sold and excludes taxes, insurance, PMI, maintenance, transaction costs, and credit constraints.

Year	10y real yield	30y mortgage	Case-Shiller index	Median sale price	Nominal median HH income	Est. 80% LTV payment	Payment / monthly nominal income
2005	1.81%	5.87%	171.74	\$236,550	\$46,330	\$1,118/mo	28.97%
2010	1.15%	4.69%	144.67	\$222,700	\$49,280	\$923/mo	22.47%
2015	0.45%	3.85%	172.16	\$294,150	\$56,520	\$1,103/mo	23.42%
2019	0.40%	3.94%	209.42	\$320,250	\$68,700	\$1,214/mo	21.20%
2020	-0.60%	3.11%	222.09	\$328,150	\$68,010	\$1,123/mo	19.81%
2021	-0.91%	2.96%	259.96	\$383,000	\$70,780	\$1,285/mo	21.78%
2022	0.43%	5.34%	298.30	\$432,950	\$74,580	\$1,933/mo	31.10%
2023	1.68%	6.81%	305.76	\$426,525	\$80,610	\$2,226/mo	33.14%
2024	1.94%	6.72%	321.37	\$418,975	\$83,730	\$2,168/mo	31.06%

The 2019–2024 comparison captures the combined effect. The 10-year TIPS real yield increased by 1.54 percentage points and the 30-year mortgage rate by 2.79 points. The median sale price increased by \$98,725. The estimated monthly principal-and-interest payment on an 80 percent loan-to-value median home increased by about \$954 per month.

V. Entry burden is a payment-income object

The most important descriptive object is not the rate or price separately. It is the payment relative to income. In the public screen used here, the estimated median-home entry payment rose sharply relative to monthly nominal median household income after 2021.

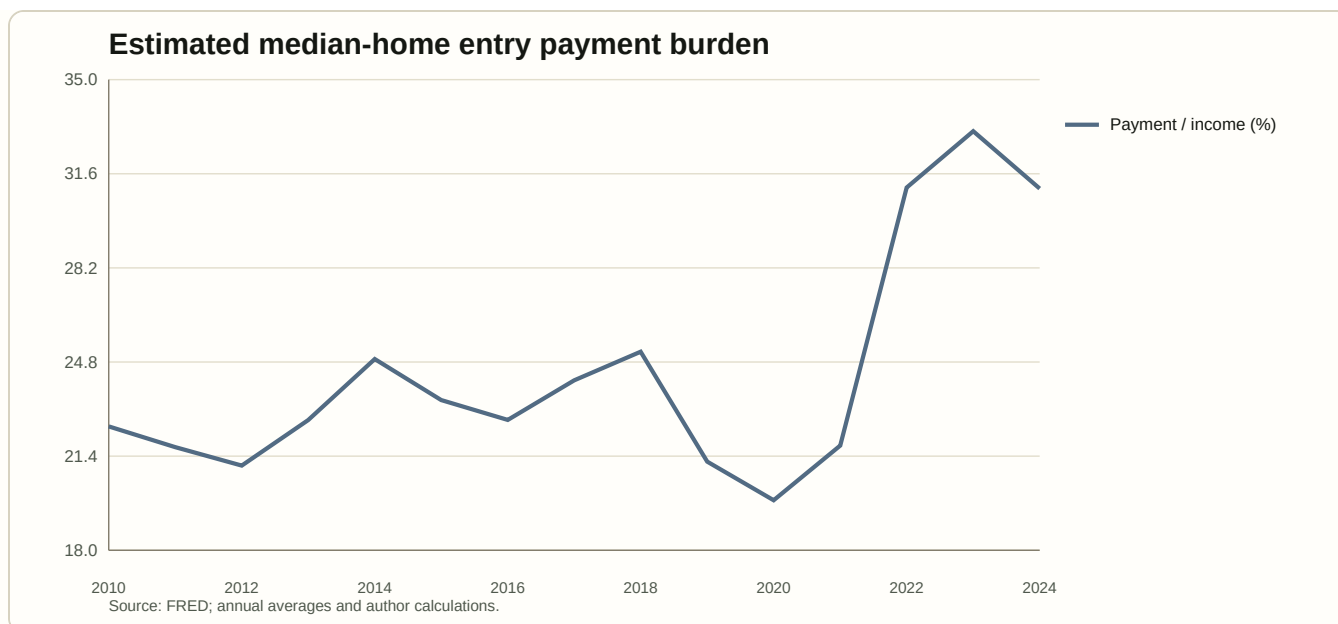


Figure 2. Estimated median-home entry payment burden.

Table 2. Regime comparisons using nominal income. Changes are differences in levels except sale price and payment, which are dollar changes. The comparison separates the low-real-rate asset-price phase from the high-financing-cost phase.

Window	Real yield change	Mortgage-rate change	Sale-price change	Est. payment change	Payment/income change
2019–2021	-1.32 pp	-0.98 pp	\$62,750	\$71/mo	0.58 pp
2021–2024	2.85 pp	3.76 pp	\$35,975	\$883/mo	9.28 pp
2019–2024	1.54 pp	2.79 pp	\$98,725	\$954/mo	9.87 pp
2020–2024	2.54 pp	3.61 pp	\$90,825	\$1,045/mo	11.26 pp

The regime comparison separates two kinds of household squeeze. The 2019–2021 window looks like a low-real-rate asset-price squeeze: median sale prices rose quickly while mortgage rates were low. The 2021–2024 window looks like a financing-cost squeeze: prices did not need to surge at the same pace for the representative entry payment to deteriorate, because mortgage rates did much of the work.

VI. Accounting decomposition and counterfactuals

The payment-burden identity allows simple accounting. Starting from the 2019 burden, updating the median sale price to its 2024 value raises the payment-income screen by 6.54 percentage points. Updating the mortgage rate to its 2024 value adds another 10.13 points. Updating nominal income to its 2024 value offsets 6.80 points. The result is a rise from 21.20 percent in 2019 to 31.06 percent in 2024.

Table 3. Accounting decomposition of the 2019–2024 entry-payment burden change.

Sequential decomposition using nominal dollars. The path starts with the 2019 burden, then updates price, mortgage rate, and income in sequence. Order matters, so this is an accounting screen rather than a causal decomposition.

Step	Price used	Mortgage rate used	Income used	Payment / monthly income	Increment
2019 baseline	\$320,250	3.94%	\$68,700	21.20%	—
Update price to 2024	\$418,975	3.94%	\$68,700	27.73%	6.54 pp
Update mortgage rate to 2024	\$418,975	6.72%	\$68,700	37.86%	10.13 pp
Update income to 2024	\$418,975	6.72%	\$83,730	31.06%	-6.80 pp

Because the sequential path is order-dependent, Table 4 reports an order-invariant version that averages each factor's marginal contribution across all six possible update orderings. The message is similar but less arbitrary: price and mortgage-rate movements both raised the burden, while nominal income growth partly offset them.

Table 4. Order-invariant accounting decomposition, 2019–2024. Shapley-style average marginal contribution across all orderings of price, mortgage rate, and income updates. Entries sum to the total 9.87 percentage-point burden increase.

Component updated from 2019 to 2024	Average contribution to burden change
Median sale price	7.00 pp
Mortgage rate	8.10 pp
Nominal household income	-5.23 pp
Total change	9.87 pp

This accounting screen is not causal. It does not say that real rates caused prices or mortgage rates to move. It says that, given observed prices, mortgage rates, and nominal income, the payment barrier worsened through both the price and financing-cost channels.

Counterfactual screens make the same point more concrete. At the observed 2024 mortgage rate, the median sale price would have needed to be roughly \$285,920 to restore the 2019 payment-to-income ratio under the same simplified 80 percent LTV setup. The observed 2024 median sale price was \$418,975.

Table 5. Counterfactual affordability screens. These are formula-based screens, not forecasts. They ask what would have been required in 2024 to restore selected 2019 benchmarks under the same simplified 80 percent LTV setup.

Counterfactual question	Required 2024 value	Observed 2024 value	Gap
Median sale price needed at 2024 mortgage rate to restore 2019 monthly payment	\$234,596	\$418,975	\$184,379 higher observed
Median sale price needed at 2024 mortgage rate/income to restore 2019 payment-income ratio	\$285,920	\$418,975	\$133,055 higher observed
Mortgage rate needed at 2024 price to restore 2019 monthly payment	1.85%	6.72%	4.87 pp higher observed
Mortgage rate needed at 2024 price/income to restore 2019 payment-income ratio	3.36%	6.72%	3.36 pp higher observed

VII. Down-payment, cash-to-close, and LTV sensitivity

The 80 percent LTV assumption is not a natural description of all entrants. It is a clean benchmark for a prime borrower with substantial cash, but many first-time buyers cannot accumulate a 20 percent down payment. Higher LTV borrowing reduces the cash down payment but raises the monthly payment and may add PMI.

Table 6 keeps the principal-and-interest screen but adds two illustrative extensions. First, it reports cash down payment plus an assumed 3 percent closing-cost allowance. This is not a complete cash-to-close model because prepaid taxes, insurance, moving costs, reserves, and lender overlays vary by household and market. Second, it reports an illustrative PITI-plus-PMI monthly burden using property taxes of 1.1 percent of price per year, homeowners insurance of 0.5 percent of price per year, and PMI of 0.5 percent of loan balance per year for LTVs above 80 percent. These are transparent approximations, not estimated local costs.

Table 6. LTV, cash-to-close, and ownership-cost sensitivity, 2024. Payment uses the 2024 median sale price and 30-year mortgage rate. PITI+PMI is illustrative and adds property taxes, insurance, and PMI assumptions described above. It still excludes maintenance, moving costs, reserves, credit constraints, and local underwriting variation.

LTV	Cash down payment	Down payment + 3% closing costs	Principal & interest	P&I / income	Illustrative PITI+PMI	Illustrative PITI+PMI / income
80%	\$83,795	\$96,364	\$2,168/mo	31.06%	\$2,726/mo	39.07%
90%	\$41,897	\$54,467	\$2,438/mo	34.95%	\$3,154/mo	45.21%
95%	\$20,949	\$33,518	\$2,574/mo	36.89%	\$3,298/mo	47.27%
97%	\$12,569	\$25,139	\$2,628/mo	37.67%	\$3,356/mo	48.10%

This sensitivity table shows why entry barriers are two-dimensional. A lower down payment can reduce the cash barrier but increase the payment barrier. A higher down payment can reduce the payment but may be impossible for liquidity-constrained renters. A complete affordability model would need both monthly payment capacity and household-specific cash-to-close capacity.

VIII. Rent and asset prices matter together

Household entry barriers are not only about buying. Rent is the alternative margin, and rent pressure can reduce the ability to accumulate down payments. Public rent CPI, home-price indexes, and earnings measures capture different objects, so the figure below rebases them to 2019 = 100 rather than comparing raw index levels.

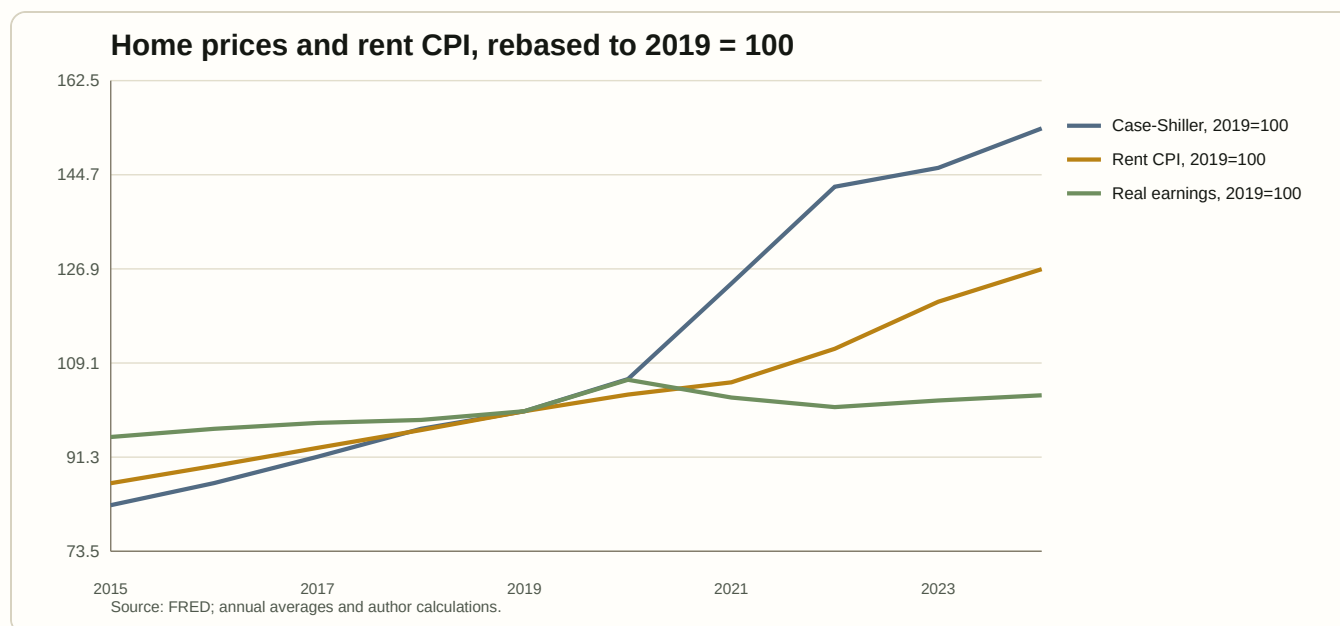


Figure 3. Home prices, rent CPI, and earnings rebased to 2019.

This figure is only motivation. Rent CPI is not asking rent, local rent burden, or residual income after rent. The paper therefore does not claim to measure renter saving capacity. A stronger empirical version would use renter-specific income, local rent levels, and a measure of how much cash a potential entrant can save after rent and essentials.

IX. Interpretation

The public data support a careful claim: higher real rates do not automatically restore household entry affordability. They may pressure asset valuations, but if prices remain elevated and mortgage rates rise, the household payment constraint can worsen.

The title emphasizes real rates because the motivation is macro-financial, but the direct household mechanism is the mortgage rate, home price, income, rent, and liquidity constraint. This note does not decompose mortgage rates into real Treasury yields, expected inflation, term premia, mortgage spreads, MBS market conditions, or credit and prepayment components. A version

focused on monetary transmission should do that decomposition. The claim here is narrower: observed higher real rates coincided with a mortgage-rate and price configuration that left the national payment screen worse, not restored.

This is not an argument for permanently low rates. Low real rates can inflate asset prices and reward incumbent owners over entrants. It is also not an argument that high rates are always bad. Higher rates can improve returns on saving, slow speculative excess, and eventually discipline valuations. The point is narrower: household entry conditions depend on the interaction of real rates, asset prices, mortgage rates, rents, income, and cash-to-close constraints.

X. Limitations and next empirical step

This note is descriptive. It does not identify causal effects of real rates on home prices, mortgage payments, rents, or household formation. It uses national aggregates and cannot separate regions, income groups, credit scores, age cohorts, renter status, or down-payment constraints.

The median sale price is not the same as the price faced by a specific first-time buyer. The median household is not the marginal entrant. The screen is incomplete even after adding illustrative ownership-cost and cash-to-close approximations. It does not measure local property taxes and insurance, maintenance, underwriting constraints, local housing supply, first-time-buyer credit access, or renter residual income.

The most natural next step is a richer local and entrant-specific accounting paper. That version would use metro-level or county-level home prices, starter-home or lower-tier prices, local incomes, renter incomes, local rents, taxes, insurance, PMI assumptions, and observed entry outcomes such as first-time-buyer mortgage originations, purchase applications, renter-to-owner transitions, household formation, or age-specific homeownership. It would then test whether places with larger increases in the payment-income burden experienced larger declines in entry outcomes.

XI. Conclusion

The question is not whether low rates or high rates are good in isolation. Low-rate regimes can raise asset prices and down-payment barriers. High-rate regimes can raise financing costs and monthly payment barriers. The household entry question is whether the total entry burden falls relative to income and cash capacity.

In the national public-data screen, higher real rates did not coincide with enough price adjustment to restore 2019-style entry-payment affordability. The result is a shift in the form of the barrier: from asset-price inflation under low real rates toward payment stress under higher rates. For a household trying to enter, the practical obstacle is not the macro regime label. It is the payment, down payment, rent alternative, cash-to-close requirement, and income margin.

Appendix A. Benchmark choice

The paper uses 2019 as the main restoration benchmark because it is the last full pre-pandemic calendar year and because the post-2021 rate increase is naturally compared with the immediately preceding low-rate expansion. This does not mean 2019 was universally affordable. Table 1 shows that the burden was lower in 2019 than in 2005, 2010, 2015, or 2024 under this national screen, but many local markets and renter groups were already constrained. The 2019 benchmark should therefore be read as a recent pre-pandemic reference point, not a normative definition of affordability.

Appendix B. Reproducibility note

The replication package contains the raw FRED CSV inputs, the processed annual panel, figure files, table outputs, and build scripts used for this working draft. Monthly, weekly, and quarterly series are converted to simple calendar-year averages. The main affordability ratio uses nominal payments divided by nominal income.

References

Federal Reserve Bank of St. Louis. 2026. *FRED Economic Data*. St. Louis, MO.

Federal Housing Finance Agency, S&P CoreLogic Case-Shiller, and U.S. Census Bureau series as distributed through FRED.